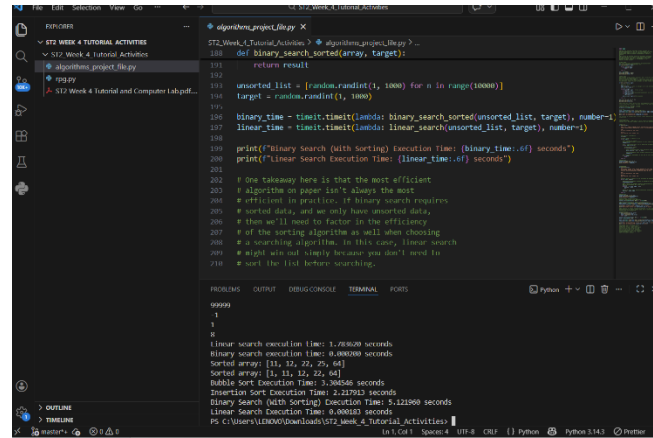


Task 1: Understanding Simple Data Structures in Python

```
File Edit Selection View Go ... ST2_Week_4_Tutorial_Activities  
ST2_Week_4_Tutorial_Activities  
  algorithms_project_file.py  
  rpg.py  
  ST2 Week 4 Tutorial and Computer Lab.pdf...  
rpg.py  
4 # going to put them to the test by creating a very basic 'game' on the console that  
5 # will leverage several of the data structures that we have learned such as dictionaries,  
6 # lists, and tuples.  
7  
8 import random as r  
9 player_health = 100  
10 enemy_health = 100  
11  
12 # Each move stores a tuple which represents changes in health to the user [0] and the tar  
13 moves = {"normal": (0, -20),  
14         "special": (5, -10),  
15         "heal": (15, 0),  
16         "last stand": (-15, -30)}  
17 moves_keys = list(moves.keys()) # so we can access keys with indices later  
18  
19 def report(player_health, enemy_health):  
20     print(f"Player health: {player_health}  
21     Enemy health: {enemy_health}  
22     ")  
23  
24 def check_game_over(player_health, enemy_health):  
25     if player_health <= 0 and enemy_health > 0:  
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```

Task 2: Understanding Simple Search Algorithms and Measuring their Computational Efficiency in Python



```
def binary_search_sorted(array, target):
    return result

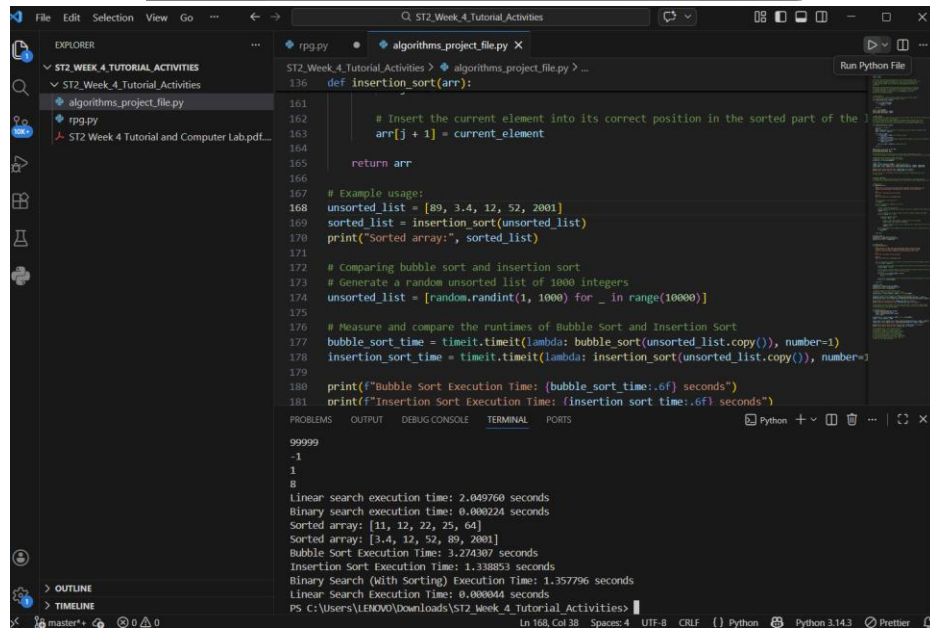
unsorted_list = [random.randint(1, 1000) for _ in range(10000)]
target = random.randint(1, 1000)

binary_time = timeit.timeit(lambda: binary_search_sorted(unsorted_list, target), number=1)
linear_time = timeit.timeit(lambda: linear_search(unsorted_list, target), number=1)

print(f"Binary Search (with Sorting) Execution Time: {binary_time:.6f} seconds")
print(f"Linear Search Execution Time: {linear_time:.6f} seconds")

# One takeaway here is that the most efficient
# algorithm on paper isn't always the most
# efficient in practice. If binary search requires
# sorted data, and we only have unsorted data,
# then we'll need to factor in the efficiency
# of the sorting algorithm as well when choosing
# a searching algorithm. In this case, linear search
# might win out simply because you don't need to
# sort the list before searching.
```

99999
-1
1
8
Linear search execution time: 1.26820 seconds
Binary search execution time: 0.00000 seconds
Sorted array: [11, 12, 22, 29, 64]
Sorted array: [11, 12, 22, 29, 64]
Bubble Sort Execution Time: 1.30556 seconds
Insertion Sort Execution Time: 2.21703 seconds
Binary Search (with Sorting) Execution Time: 0.12196 seconds
Linear Search Execution Time: 0.00000 seconds
PS C:\Users\LENOVO\Downloads\ST2_Week_4_Tutorial_Activities>



```
def insertion_sort(arr):
    # Insert the current element into its correct position in the sorted part of the list
    arr[j + 1] = current_element
    return arr

# Example usage:
unsorted_list = [89, 3, 4, 12, 52, 2001]
sorted_list = insertion_sort(unsorted_list)
print("Sorted array:", sorted_list)

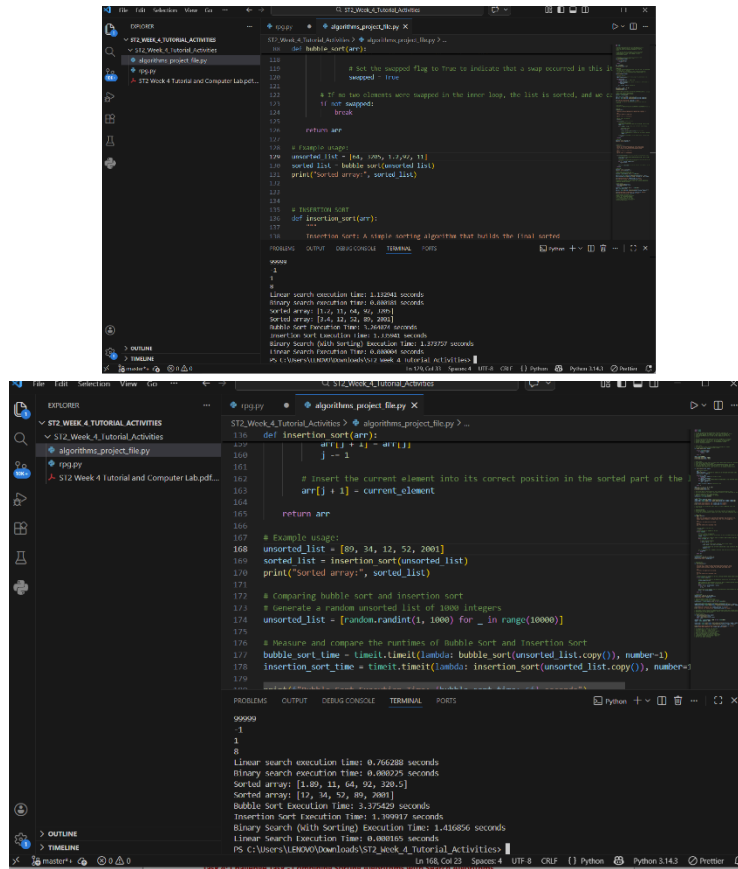
# Comparing bubble sort and insertion sort
# Generate a random unsorted list of 1000 integers
unsorted_list = [random.randint(1, 1000) for _ in range(10000)]

# Measure and compare the runtimes of Bubble Sort and Insertion Sort
bubble_sort_time = timeit.timeit(lambda: bubble_sort(unsorted_list.copy()), number=1)
insertion_sort_time = timeit.timeit(lambda: insertion_sort(unsorted_list.copy()), number=1)

print(f"Bubble Sort Execution Time: {bubble_sort_time:.6f} seconds")
print(f"Insertion Sort Execution Time: {insertion_sort_time:.6f} seconds")
```

99999
-1
1
8
Linear search execution time: 2.049760 seconds
Binary search execution time: 0.000224 seconds
Sorted array: [11, 12, 22, 29, 64]
Sorted array: [3, 4, 12, 52, 89, 2001]
Bubble Sort Execution Time: 3.274307 seconds
Insertion Sort Execution Time: 1.338853 seconds
Binary Search (with Sorting) Execution Time: 1.357796 seconds
Linear Search Execution Time: 0.000044 seconds
PS C:\Users\LENOVO\Downloads\ST2_Week_4_Tutorial_Activities>

Task 3: Understanding Simple Sorting Algorithms and Measuring their Computational Efficiency in Python



The image displays two screenshots of a Python IDE, likely VS Code, showing the implementation and execution of sorting algorithms.

Top Screenshot: The code defines a `bubble_sort` function. It includes a `swapped` flag to optimize the process by stopping if no swaps occur in a pass. The function is tested with an example array `[14, 34, 12, 52, 2001]` and a random array of 1000 integers. The terminal output shows the execution times for Linear search, Binary search, Bubble sort, and Insertion sort.

```
def bubble_sort(arr):
    n = len(arr)
    # Traverse through all elements in the array
    for i in range(n):
        # Set the swapped flag to True in the first loop
        swapped = True
        # Traverse the array from 0 to n-i-1
        for j in range(1, n-i):
            # If no two elements were swapped in the inner loop, the list is sorted, and we can break
            if arr[j] > arr[j+1]:
                arr[j], arr[j+1] = arr[j+1], arr[j]
                swapped = False
        # If no two elements were swapped in the inner loop, the list is sorted, and we can break
        if not swapped:
            break
    return arr

# Example usage
unsorted_list = [14, 34, 12, 52, 2001]
sorted_list = bubble_sort(unsorted_list)
print("Sorted array:", sorted_list)

# Insertion Sort
def insertion_sort(arr):
    # Insertion Sort: A simple sorting algorithm that builds the final sorted
    # array one item at a time.
    for i in range(1, len(arr)):
        current_element = arr[i]
        position = i
        while position > 0 and arr[position-1] > current_element:
            arr[position] = arr[position-1]
            position = position - 1
        arr[position] = current_element
    return arr

# Example usage
unsorted_list = [89, 34, 12, 52, 2001]
sorted_list = insertion_sort(unsorted_list)
print("Sorted array:", sorted_list)

# Comparing bubble sort and insertion sort
# Generate a random unsorted list of 1000 integers
unsorted_list = [random.randint(1, 1000) for _ in range(1000)]

# Measure and compare the runtimes of Bubble sort and Insertion sort
bubble_sort_time = timeit.timeit(lambda: bubble_sort(unsorted_list.copy()), number=1)
insertion_sort_time = timeit.timeit(lambda: insertion_sort(unsorted_list.copy()), number=1)

print("Bubble Sort Execution Time: {}".format(bubble_sort_time))
print("Insertion Sort Execution Time: {}".format(insertion_sort_time))
```

Bottom Screenshot: This screenshot shows the same code as the top one, but with the terminal output displaying the execution times for the sorting algorithms. The output shows that Bubble sort is significantly faster than Insertion sort for the given data.

```
Linear search execution time: 1.11041 seconds
Binary search execution time: 0.00000 seconds
Sorted array: [14, 12, 34, 52, 2001]
Sorted array: [14, 12, 34, 52, 2001]
Bubble sort execution time: 1.36429 seconds
Insertion sort execution time: 1.36429 seconds
Binary Search (with Sorting) execution time: 1.36429 seconds
Linear Search execution time: 0.00000 seconds
```

Task 4: Challenge Task - Combining Sorting Algorithms with Search Algorithms

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\LENOVO\AppData\Local\Programs\Microsoft VS Code> & C:\Users\LENOVO\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/LENOVO/Desktop/ST2_Week_4_Tutorial_Activities (1)/ST2_Week_4_Tutorial_Activities/algorithms_project_file.py"
```

```
=====
```

```
DATA SIZE = 1000
```

```
=====
```

```
Insertion Sort + Linear Search: 0.019800 sec  
Bubble Sort + Linear Search:    0.057268 sec  
Insertion Sort + Binary Search: 0.019488 sec  
Bubble Sort + Binary Search:    0.044860 sec
```

```
=====
```

```
DATA SIZE = 3000
```

```
=====
```

```
Insertion Sort + Linear Search: 0.195801 sec  
Bubble Sort + Linear Search:    0.454742 sec  
Insertion Sort + Binary Search: 0.199373 sec  
Bubble Sort + Binary Search:    0.808435 sec
```

```
=====
```

```
DATA SIZE = 5000
```

```
=====
```

```
Insertion Sort + Linear Search: 0.675477 sec  
Bubble Sort + Linear Search:    1.474553 sec  
Insertion Sort + Binary Search: 0.629988 sec  
Bubble Sort + Binary Search:    1.285437 sec
```

```
PS C:\Users\LENOVO\AppData\Local\Programs\Microsoft VS Code> █
```